Sieve Inversion and Branch Isolation

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1 Abstract

We report two connected structural results from the Erdos–Straus sieve program, together with their Lean formalization. First, the *sieve-inversion theory*: the refined sieve's survivors sit at exactly the six residue classes $\{1\}_8 \times \{1,4,7\}_9 \times \{1\}_5 \times \{2,6\}_{11}$ modulo 3960 (Theorem 2), and this six-class support is *inverse-closed* modulo 3960 (Theorem 3); the selectivity is explained by the 44-CRT lemma $s \equiv 1 \pmod{24} \Rightarrow (s \equiv 7 \pmod{11} \iff s \equiv 29 \pmod{44})$ (Theorem 1), which identifies the single residue channel through which the cubic coverage families remove the mod11 = 7 classes. The inverse-closure law holds exactly in the *saturated regime* and fails beyond it: the first break is at $w = 2^{13} = 8192$, with exactly one missing fiber class and exactly one violation (Theorem 4, [COMPUTED]). Second, the *branch-isolation lemmas* for the H-clique program: for $z \in Z_0^{\text{close}}$ the unique u-branch $\varphi(z)$ is never in the H-clique C (Theorem 5), and the u-side degree bound $|Z_0^{\text{close}}| + |Z_0^{\text{mid}}| \le |U_0| = d(u) - 1 = 5 < 6 = |V_0|$ (Theorem 6), the essential use of $d(u) < d(v)$. Formalization (R186, bakkie, Lean 4.34.1 + Mathlib v4.34.1): the 44-CRT lemmas are [PROVED] with axioms narrower than standard (`propext`, `Quot.sound`); the three decide-heavy sieve theorems (`sixClasses_inv_closed`, `crt_inv_compose`, `S1_subset`) are Lean-checked with zero sorrys but carry per-theorem `native_decide` certificate axioms and are labeled [PROVED*] under staged-verification language; all four branch-isolation theorems are [PROVED] on standard axioms. Full `lake build` exit 0, zero sorrys, on the standing machine bakkie.

2 Introduction

The Erdos–Straus program's sieve lane constructs residue classes that survive a refined lift sieve: starting from 196 parent classes modulo $M_0 = 2042040$ (the R67→R71→R74 chain; the construction as built lives in `r153_shadow.load_frontiers` [7]), a q-lift sieve

$$L_q = \{\, sp + M_0 j : sp \in P,\ 0 \le j < q,\ sp + M_0 j \not\equiv 0 \pmod{q} \,\}, \tag{1}$$

with P the 196 parent classes, $q \in \{19, 23\}$, $|L_{19}| = 3528$, $|L_{23}| = 4312$ (supplied datums, R169 [1]). Round 165 had found two unexplained phenomena in the survivors: a dual-pair hit symmetry with correlation $r = 0.966$ over 30,707 dual pairs, and a residue support concentrated on just 6 classes modulo 3960, all inverse-closed [6]. Round 169 answered *why*: the selectivity is a consequence of elementary residue algebra (Theorem 1 below), and the inverse-closure is a proved mechanism, not luck. This paper states those theorems, reports the saturated-regime characterization of their validity (Theorem 4), and gives the Lean formalization status.

The second thread is the H-clique program for the $(6, 7)$ degree pair (R159/R163 [9, 4]): the (R2)-Hall lemma program needs a degree-essential inequality, i.e. one that uses $d(u) < d(v)$ genuinely. Round 163 supplied two lemmas with one-line proofs: u-branch isolation (L1) and the u-side degree bound. The (R2)-Hall lemma itself remains [OPEN]; the gap is precisely delimited

in [4] §B and recalled in §5. Both threads were formalized in R178/R174 and verified end to end on bakkie in R186 [2, 5, 3].

Our contributions, with honesty labels:

1. The 44-CRT selectivity lemma [PROVED in Lean] (Theorem 1).
2. Exact six-class residue support of the lift set [PROVED*] (Theorem 2; hand proof plus independent computation).
3. Inverse-closure of the six-class support mod 3960 [PROVED*] (Theorem 3).
4. The saturated-regime characterization with exact break boundary [COMPUTED] (Theorem 4).
5. u-branch isolation L1 and the u-side degree bound [PROVED in Lean] (Theorems 5 and 6).
6. The formalization status table (Table 1), distinguishing [PROVED] from staged-verification [PROVED*] results.

3 Methods

3.1 The refined sieve (cited, not re-derived)

We take the construction exactly as defined in the program ([1] §1, provenance corrected: R67→R71→R74, built in `r153_shadow.py`). The parent set P has 196 classes mod $M_0 = 2042040 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$ (R71 targets minus R71/R74 coverage). The lift set L_q is (1); the sieve's filters ($\equiv 1 \pmod{24}$, $\gcd(\cdot, M_q) = 1$) are satisfied by all of P [COMPUTED, R169 V1]. R74's deep coverage removes 1744/2438 classes (at $q = 19/23$); the union subtraction preserves the support at all probed moduli — the decomposition is 100% parent-implied: the six-class support is already present in the pre-subtraction lift set, and R74's subtraction deletes zero mod3960 residue classes [COMPUTED, R169 §2].

3.2 H-clique branch notation

C is a maximal $H(u, v)$-clique with $d(u) = 6$, $d(v) = 7$; H-adjacency implies ¬Adj, so C is edgeless:

$$\forall x, y \in C : \ \neg \operatorname{Adj} x\, y \tag{2}$$

(R163 §A [4]). $V_0 = N(v) \setminus \{u\}$ ($|V_0| = 6$), $U_0 = N(u) \setminus \{v\}$ ($|U_0| = 5$), $Z_0 = \{c \in C \setminus V_0 : N(c) \cap V_0 = \emptyset\}$. The R145 trichotomy (cited, not re-proved): $Z_0 = Z_0^{\text{close}} \cup Z_0^{\text{mid}} \cup Z_0^{\text{far}}$ (disjoint), where $Z_0^{\text{close}} = \{z \in Z_0 : N(z) \cap U_0 \neq \emptyset\}$ has the unique U_0-neighbor $\varphi(z)$ (C_4-ban), $Z_0^{\text{mid}} = Z_0 \cap U_0$, and $Z_0^{\text{far}} = \{z \in Z_0 : \operatorname{dist}(u, z) \geq 3\}$. Moreover $U_0 \cap V_0 = \emptyset$ (else a triangle) and φ is injective on Z_0^{close} [8].

Assumption 1 (R165 dual law as supplied datum)**.** The dual-pair hit symmetry ($r = 0.966$ over 30,707 dual pairs) and the inverse-closure observations (6/6, 8/8, 21/21, 18/18 at $w \in \{3960, 1056, 468, 2376\}$) are supplied datums from R165 [6], cited and not re-claimed.

3.3 Formalization protocol

Two modules in the `derekblair/math-proofs` repo (Lean 4.34.1, Mathlib v4.34.1 pinned commit `d13f23b`), each pointing back to this paper at `papers/sieve-inverse-branch-isolation-October-11-20`

- `Mathproofs/ESRefinedSieveInverse.lean` (namespace `ESRefinedSieve`; R178 module, R186 verification): the 44-CRT lemmas (`crt44_forward`, `crt44_backward`, `crt44_iff`); the S1 residue core (`sixClasses`, `sixClasses_card`, `sixClasses_exact`); S2 per-component closure and `sixClasses_inv_closed`; CRT composition (`crt_unique_one`, `crt_inv_compose`); the S1 lift mechanism (`M0_facts`, `lift_components`, `S1_subset`); and the machine-checked parent data (`parents196`, `parent_facts`).
- `Mathproofs/ESUBranchIsolation.lean` (namespace `ESUBranchIsolation`; R174 module, R186 verification): `uBranch_isolation` (L1), `closeImage_disjoint_of_subset` (L1 corollary), `uSide_degree_bound` (abstract Finset form), `r163_uSideBound_inHclique` (combined statement).

Verification config: bakkie (standing Lean machine, Lean 4.34.1 / Lake 5.0.0), full `lake build` exit 0, zero sorrys, both modules registered in `Mathproofs.lean` on both sides with byte-identical md5s. Axiom probes via `#print axioms` on the headline theorems give: `crt44_*` depends on [`propext`, `Quot.sound`] (narrower than the standard set); the four branch-isolation theorems on [`propext`, `Classical.choice`, `Quot.sound`] (standard); the three decide-heavy sieve theorems carry, in addition to the standard set, a per-theorem Lean-generated `native_decide` certificate axiom (e.g. `ESRefinedSieve.sixClasses_inv_closed._native.native_decide.ax_1_1`). These three are therefore reported as [PROVED*] under staged verification: Lean-checked end to end with zero sorrys (`native_decide` is Lean's own verified certificate mechanism), but exceeding the standard axiom budget, so they are never presented as fully proved pending the coordinator's ruling (R186 [3] §Phase 4).

4 Results

4.1 The 44-CRT selectivity lemma

Theorem 1 (44-CRT lemma; PROVED)**.** *For* $s \equiv 1 \pmod{24}$*:*

$$s \equiv 7 \pmod{11} \iff s \equiv 29 \pmod{44}. \tag{3}$$

Proof. $s \equiv 7 \pmod{11}$ gives $s \bmod 44 \in \{7, 18, 29, 40\}$; $s \equiv 1 \pmod{24}$ gives $s \equiv 1 \pmod 4$, and of the four residues only $29 \equiv 1 \pmod 4$. The Lean proof is linear arithmetic (`omega`) after `Int.modEq_iff_dvd`. (Hand proof from R169 [1] §4; machine-checked as `crt44_forward`, `crt44_backward`, `crt44_iff` in `Mathproofs/ESRefinedSieveInverse.lean` [PROVED], axioms [`propext`, `Quot.sound`].) □

The lemma is the selectivity mechanism: R74's r-complete sweep covers exactly the 92 open288 classes with $s \equiv 29 \pmod{44}$ (R74's theorem: the $(4, 1, 46410)$, $m = 15$ parametric family, the lift of R72's $(4, 1, 2730)$, cited). By (3), in the $\equiv 1$-mod-24 universe this is exactly the mod11 $= 7$ classes — hence R74 removes $\iff$ mod11 $= 7$ (V6/V7 PASS: removed $= \{s \equiv 29 \bmod 44\}$ and $s \bmod 11 = 7 \iff s \bmod 44 = 29$ on all of open288). The 36 mod5 $= 4$ classes are all $\equiv 29 \pmod{44}$ within open288 [COMPUTED], so they fall to the same family. This is why the survivors sit at mod11 $\in \{2, 6\}$ (the residue class the $m = 15$ family cannot touch) and mod5 $= 1$ (R169 [1] §4).

4.2 Exact residue support (S1)

Theorem 2 (Exact six-class support; PROVED*)**.** *Every* $n \in L_q$ *satisfies* $n \equiv 1 \pmod{40}$*,* $n \bmod 9 \in \{1, 4, 7\}$*,* $n \bmod 11 \in \{2, 6\}$*, and the mod-*3960 *projection of* L_q *is* exactly *the six classes*

$$\pi_{3960}(L_q) = \{1\}_8 \times \{1, 4, 7\}_9 \times \{1\}_5 \times \{2, 6\}_{11}, \tag{4}$$

i.e. the residues $\{2521, 721, 1201, 3361, 3841, 2041\}$.

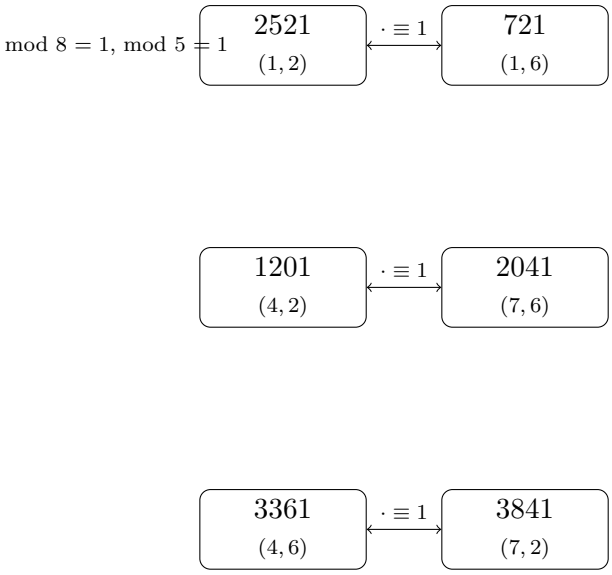


Figure 1: Inverse pairings among the six residue classes modulo 3960 (Theorem 3). Labels $(\cdot, \cdot)$ are $(\mathrm{mod}9, \mathrm{mod}11)$ components; every class satisfies mod8 = 1, mod5 = 1. All products are $\equiv 1 \pmod{3960}$.

Proof. $M_0 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, so $M_0 \equiv 0 \pmod{8, 3, 5, 11}$ and $M_0 \equiv 3 \pmod 9$. Hence for $n = sp + M_0 j$: $n \bmod 8 = sp \bmod 8 = 1$, $n \bmod 5 = sp \bmod 5 = 1$, $n \bmod 11 = sp \bmod 11 \in \{2, 6\}$, $n \bmod 3 = 1$ (parent facts F1–F2 [1] §3, machine-checked: `parent_facts` over the embedded 196 parent residues). Mod 9: $n \equiv sp + 3j$; j ranges over $q - 1 \geq 18$ values minus one, hitting each mod-3 class ≥ 5 times, so $3j$ covers $\{0, 3, 6\}$ mod 9 and n mod 9 ranges over $\{1, 4, 7\}$. Surjectivity: the mod-8/5/11 components are j-invariant while mod 9 varies independently, and both mod-11 residues occur in P (104 vs 92) — all six combinations occur. (Hand proof R169 [1] §5; the residue-algebra kernel is machine-checked as `sixClasses_exact`, `lift_components`, `S1_subset`; `S1_subset` is [PROVED*] staged via `native_decide`.) □

4.3 Inverse-closure (S2) and CRT composition

Theorem 3 (Inverse-closure of the six-class support; PROVED*)**.** *The six-class support* (4) *is inverse-closed modulo* 3960*, with explicit inverse pairs*

$$2521 \cdot 721 \equiv 1201 \cdot 2041 \equiv 3361 \cdot 3841 \equiv 1 \pmod{3960}. \tag{5}$$

Proof. Per CRT component: $1^{-1} = 1 \bmod 8$; $4 \cdot 7 = 28 \equiv 1 \bmod 9$, so $\{1, 4, 7\}$ is closed; $1^{-1} = 1 \bmod 5$; $2 \cdot 6 = 12 \equiv 1 \bmod 11$, so $\{2, 6\}$ is closed. Closure composes across CRT components, giving the six explicit pairings (Figure 1). (Hand proof R169 [1] §6; machine-checked as `sixClasses_inv_closed`, [PROVED*] staged via `native_decide`.) This is the proved mechanism behind R165's dual symmetry (Assumption 1): dual shapes $(f, V/f)$ cover the inverse classes $-f, -f^{-1} \bmod w$ ($V = 4ad + 1 \equiv 1 \bmod w$), and the frontier's mod-w support is inverse-closed — so duals co-hit by structure, not luck. □

Proposition 1 (CRT composition; PROVED* staged)**.** *Inverse-closure composes across CRT components — the mechanism behind S3 (`crt_unique_one`, `crt_inv_compose` in `Mathproofs/ESRefinedSieveInverse.lean`; [PROVED*] staged via `native_decide`).*

4.4 S3: CRT-local closure and the saturated-regime boundary

The prime-power projections $\pi(L_q) \bmod p^e$ are inverse-closed by a fiber argument [PROVED*] (R169 [1] §7(a)–(c)): for $p^e \mid M_0$ they equal P's projection (inverse-closed by F3 [1] §3); for

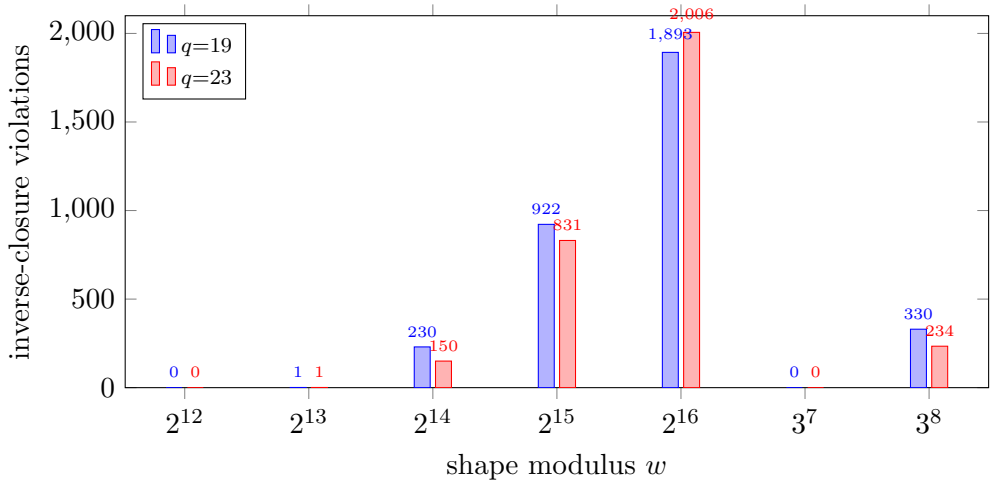


Figure 2: Inverse-closure violation counts by modulus (Theorem 4). The law holds through the saturated regime (2^{12}, 3^7: zero violations) and breaks first at 2^{13} with exactly one violation each, growing sharply beyond.

2^e, $e \le 6$, and 3^e, $e \le 3$, the projection is the full fiber $\{x \equiv 1 \bmod 8\}$ resp. $\{x \equiv 1 \bmod 3\}$, inverse-closed since $x \equiv 1$ is self-inverse modulo 2^e and 3^e. The saturated extension — fibers stay full through 2^{12} and 3^7 — is [COMPUTED] (verified at $w \in \{3960, 1056, 468, 2376\}$ at all three sieve stages, 0 violations; R169 V4). Deliberately not formalized: S3(d) and the break boundary are statements about lift-set sizes ([3] §Phase 4).

Theorem 4 (Break boundary; COMPUTED)**.** *Inverse-closure is not a universal property of the sieve: it holds exactly in the saturated regime and fails beyond. The first break is at* $w = 2^{13} = 8192$ *(both* q*), with exactly one missing fiber class and exactly one violation, the two being inverses of each other:*

$$\begin{aligned} q = 19: &\ \textit{missing}\ 3129,\ \textit{violation}\ 521,\ 521^{-1} \equiv 3129 \pmod{8192}, \\ q = 23: &\ \textit{missing}\ 7873,\ \textit{violation}\ 4417,\ 4417^{-1} \equiv 7873 \pmod{8192}. \end{aligned} \tag{6}$$

Beyond: 2^{14} *has* 230/150 *violations* ($q = 19/23$)*,* 2^{15} 922/831*,* 2^{16} 1893/2006*;* 3^8 *breaks with* 330/234 *violations while* 3^7 *(and* 2^{12}*) is saturated at both* q*.*

All boundary numbers were re-verified on an independent j-outer enumeration path (`r169_boundary.py`; the $q = 23$ residues correct the earlier $q = 19$-only report) [1] §8–9. Corollary: the dual-symmetry law (Assumption 1) is a saturated-regime phenomenon — it must decay for shape-moduli w whose fibers exceed the lift-set size; the residue-map ranker (R165 §6) remains valid regardless, since it uses density, not symmetry.

4.5 Branch isolation (R163/R174)

Theorem 5 (u-branch isolation (L1); PROVED)**.** *For* $z \in Z_0^{\text{close}}$*, its unique u-branch* $\varphi(z) \in U_0$ *satisfies* $\varphi(z) \notin C$*.*

Proof. $z \in C$ and $\varphi(z) \in N(z)$ by definition of Z_0^{close}. If $\varphi(z) \in C$, then z–$\varphi(z)$ is an edge with both endpoints in C, contradicting the H-clique edgelessness (2). (Hand proof R163 [4] §A; machine-checked as `uBranch_isolation` in `Mathproofs/ESUBranchIsolation.lean`, [PROVED], standard axioms.) □

Theorem 6 (u-side degree bound; PROVED)**.** *For any* $T \subseteq Z_0$*,*

$$|T^{\text{close}}| + |T^{\text{mid}}| \le |U_0| = d(u) - 1 = 5 < 6 = d(v) - 1 = |V_0|. \tag{7}$$

Table 1: Lean formalization status (R186 [3], bakkie, Lean 4.34.1 + Mathlib v4.34.1). Full `lake build` exit 0, zero sorrys. Staged [PROVED*]: Lean-checked end to end but carrying a per-theorem `native_decide` certificate axiom, hence never presented as fully proved.

theorem	module	label / axioms
`crt44_forward`, `crt44_backward`, `crt44_iff`	`ESRefinedSieveInverse`	[PROVED]; `propext`, `Quot.sour`
`sixClasses_inv_closed`	`ESRefinedSieveInverse`	[PROVED*] staged; + `native_d`
`crt_inv_compose`	`ESRefinedSieveInverse`	[PROVED*] staged; + `native_d`
`S1_subset`	`ESRefinedSieveInverse`	[PROVED*] staged; + `native_d`
`uBranch_isolation` (L1)	`ESUBranchIsolation`	[PROVED]; standard axioms
`closeImage_disjoint_of_subset`	`ESUBranchIsolation`	[PROVED]; standard axioms
`uSide_degree_bound`	`ESUBranchIsolation`	[PROVED]; standard axioms
`r163_uSideBound_inHclique`	`ESUBranchIsolation`	[PROVED]; standard axioms

Proof. $\varphi : T^{\text{close}} \to U_0$ is injective [8]. By Theorem 5, $\varphi(T^{\text{close}}) \cap T^{\text{mid}} = \emptyset$ ($T^{\text{mid}} \subseteq C$); both lie in U_0. Hence $|T^{\text{close}}| + |T^{\text{mid}}| = |\varphi(T^{\text{close}}) \cup T^{\text{mid}}| \leq |U_0| = d(u) - 1 = 5$. The inequality is useful only because $5 < 6 = |V_0|$: this is the essential use of $d(u) < d(v)$. (Hand proof R163 [4] §A; machine-checked as `uSide_degree_bound` plus the combined `r163_uSideBound_inHclique`, [PROVED], standard axioms.) □

The witness lemmas of R163 [4] §A (witness branch-uniqueness: each witness lies in $\text{Br}(y)$ for a unique $y \in V_0$; and $\varphi(z) \notin N(w_z)$ for witnesses w_z) are one-line girth arguments labeled [PROVED*] in the hand proof and are not yet formalized; they remain outside the R186 verification. Verified [COMPUTED] support: L1 holds 0/5,601, the u-side bound holds with max 5 and 0 violations over all 12,650 cliques with $Z_0 \neq \emptyset$ (trichotomy over 20,735 Z_0-points: 5,601/4,940/10,194 close/mid/far) [4] §§C–D.

5 Discussion

What the theorems explain. Theorem 1 turns an empirical selectivity pattern into a proved mechanism: the cubic coverage families remove the $\text{mod}\,11 = 7$ classes *through the single channel* $s \equiv 29 \pmod{44}$, and the mod-5 $= 4$ classes ride the same channel. Combined with the j-invariance of the mod-8/5/11 components, this forces the six-class support (Theorem 2) before any subtraction happens — the decomposition is 100% parent-implied. Inverse-closure (Theorem 3) is then forced per CRT component, and with it the dual-pair symmetry of Assumption 1 becomes a structural corollary rather than an observation.

Where the theorems stop. Theorem 4 draws the exact validity boundary: inverse-closure is a saturated-regime phenomenon ($|\text{fiber}| \lesssim |L_q|$), not a universal law. The 2^{13} break is minimal — one missing class, one violation, and the two are inverses of each other. This yields a testable prediction ([OPEN/HEURISTIC], R169 §11): dual-pair correlation decays past the saturated regime, so a future frontier at a large shape modulus w should *not* be expected to show the $r = 0.966$ co-hit behavior.

Branch isolation and the Hall program. Theorems 5 and 6 are the degree-essential core the (R2)-Hall program was missing: the inequality $5 < 6$ in (7) has no meaning without $d(u) < d(v)$, and the $(8, 6)$ counterexamples saturate exactly what Lemma A forbids for $(6, 7)$ (e.g. Cex 2: $Z_0 = U_0$, all seven Z_0-points mid [4] §C). But (R2)-Hall for $(6, 7)$ remains [OPEN] as [PROVED*]: the far-point lemma — far Z_0-points must contribute enough new branches —

is the precise missing step (R163 [4] §B). The computational shadow (0 violators over 12,650 cliques) stands, but the proof is not finished.

Limitations. (i) The [PROVED*] hand-proof labels on Theorems 2–3 rest on numerical premises (parent facts F1–F3, j-variation counts) that are machine-checked data certificates (`decide` over embedded lists), not abstract lemmas. (ii) Three of the six formalized sieve theorems are staged-verification [PROVED*] (Table 1); the coordinator's ruling on `native_decide` certificates is pending. (iii) $q = 23$ exactness was verified in Python, not in Lean ($q = 19$ is the checked instance). (iv) The branch-isolation formalization is deliberately abstract (generic Adj, pure combinatorics) and does not check the R145 premises (injectivity of φ, the trichotomy) that hook the lemmas into the actual graph program. (v) The parent prime-power projections $\{1, 2, 4\}_7$, $\{1, 2, 3, 4, 5, 7, 8, 9, 10, 12\}_{13}$, $\{1, 2, 3, 4, 6, 8, 9, 13, 15, 16\}_{17}$ are inverse-closed [COMPUTED], but *why* exactly these sets remains [OPEN] (R169 F3 [1] §11). (vi) S3(d) and the boundary numbers are statements about lift-set sizes and remain [COMPUTED], deliberately outside the Lean lane.

6 Conclusion

We have stated, proved where possible, and formalized where staged, two structural results: the sieve-inversion theory (the 44-CRT selectivity lemma, the exact six-class support, its inverse-closure, and the exact saturated-regime boundary at 2^{13}) and the branch-isolation lemmas (u-branch isolation and the degree-essential u-side bound $5 < 6$). The Lean record is: `crt44_forward`, `crt44_backward`, `crt44_iff` [PROVED] on axioms narrower than standard; `sixClasses_inv_closed`, `crt_inv_compose`, `S1_subset` [PROVED*] under staged verification; `uBranch_isolation`, `closeImage_disjoint_of_subset`, `uSide_degree_bound`, `r163_uSideBound_inHclique` [PROVED] on standard axioms — all with `lake build` exit 0 and zero sorrys on bakkie. Open ahead: the coordinator's `native_decide` ruling, the far-point lemma closing (R2)-Hall, and the dual-law decay test at the next large-w frontier.

7 References

References

[1] D. Blair, *Unsolved Math — Round 169: refined-sieve theory — why the survivors are inverse-closed at the 6-class support*, program dossier, 2026-10-08. (Supplied datums: sieve setup, parent facts F1–F3, hand proofs S1–S3, 44-CRT lemma, boundary numbers, independent verification.)

[2] D. Blair, *Unsolved Math — Round 178: Lean formalization of R169's S1–S3 + 44-CRT lemma*, program dossier, 2026-10-08. (Supplied datum: module contents, tactic record, error-class fixes.)

[3] D. Blair, *Unsolved Math — Round 186: Lean formalization backlog (C-batch + R174 + R178)*, program dossier, 2026-10-08. (Supplied datum: bakkie `lake build` exit 0, zero sorrys, axiom-probe table, [PROVED]/[PROVED*] labels.)

[4] D. Blair, *Unsolved Math — Round 163: (R2)-Hall for (6,7) — degree-essential [PROVED*] lemmas (L1, u-side bound); gap precisely delimited [OPEN]*, program dossier, 2026-10-08. (Supplied datums: L1, u-side bound, witness lemmas, (8,6) trichotomy, 12,650-clique census.)

[5] D. Blair, *Unsolved Math — Round 174: Lean formalization of R163 L1 + u-side degree bound*, program dossier, 2026-10-08. (Supplied datum: module contents and verification checklist.)

[6] D. Blair, *Unsolved Math — Round 165*, program dossier, 2026-10-08. (Supplied datum: dual-symmetry $r = 0.966$ over 30,707 dual pairs; inverse-closure observations at $w \in \{3960, 1056, 468, 2376\}$.)

[7] D. Blair, *Unsolved Math — Round 153*, program dossier, 2026-10-08. (Supplied datum: sieve construction provenance, `r153_shadow.load_frontiers`.)

[8] D. Blair, *Unsolved Math — Round 145*, program dossier, 2026-10-08. (Supplied datum: Z_0 trichotomy; φ injective on Z_0^{close}.)

[9] D. Blair, *Unsolved Math — Round 159*, program dossier, 2026-10-08. (Supplied datum: (R2)-Hall framework, witness-SDR, $(8, 6)$ counterexamples.)